AI OPEN
DATA CENTER

Commissioning Liquid Cooled Data Centers With Computing Equipment's for Operational Readiness

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Panel Discussion



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Why Commissioning Cannot End at IST Handover

TRADITIONAL VERIFICATION

Load Banks / IST

Static resistive load
Predictable heat input
Steady-state capacity
Integrated system operation

Answers:

“Can the plant support the design load?”

THE VALIDATION GAP

Test Conditions ≠ Production AI Behavior

LEVEL 5b

Performance Verification and Validation

Real compute load → transient response → setpoint validation → measured operating envelope

REAL WORKLOAD VALIDATION

Actual Compute Load

Transient GPU/TPU demand
Training + inference variability
Real thermal response
Real control-system interaction

Answers:

“How does the plant actually respond?”

COMMISSIONING LIFECYCLE

Level 5a IST
Process and System Startup Commissioning

→

Level 5b
Performance Verification and Validation

→

Level 5c
Simulated Failures

→

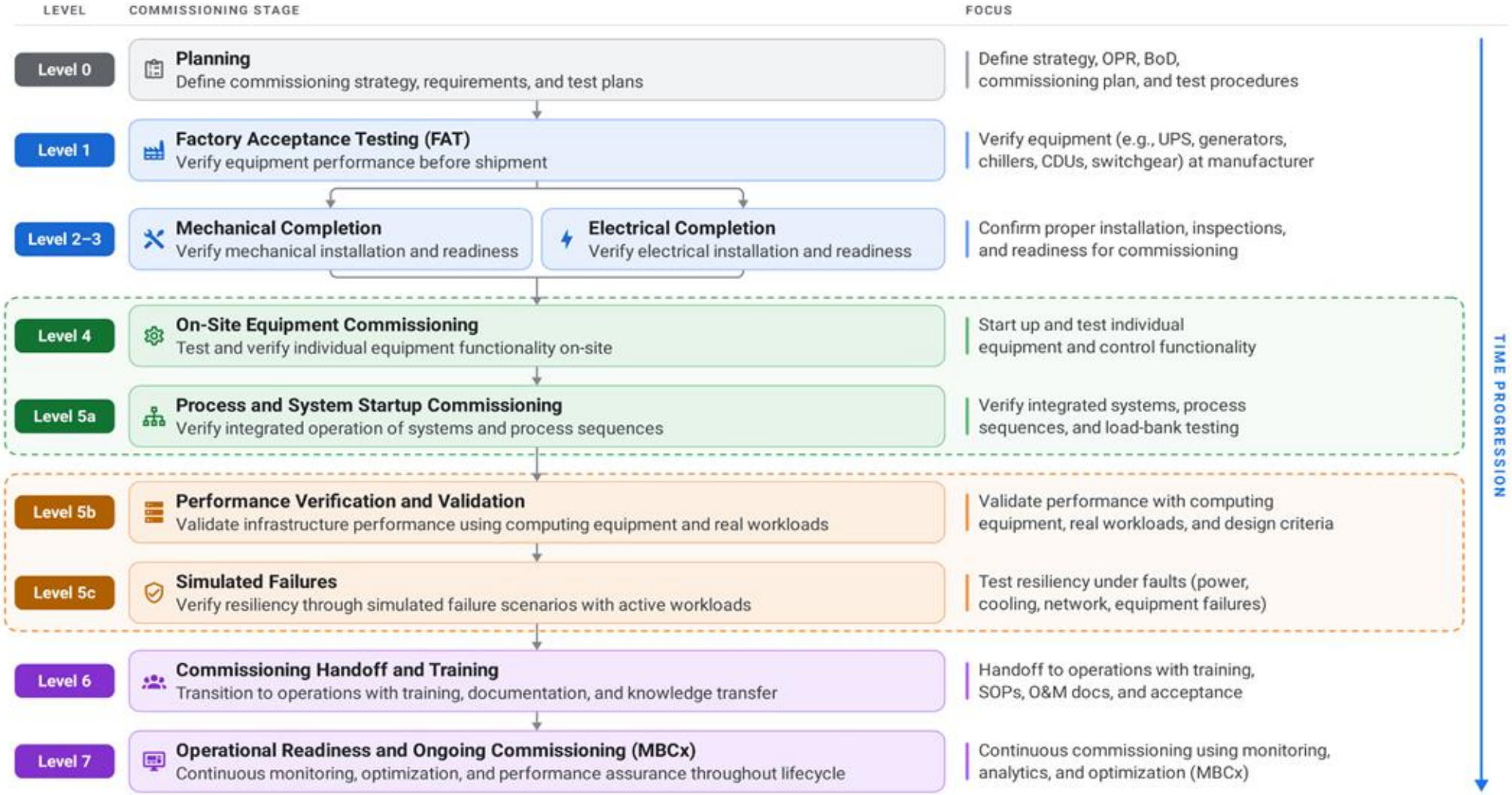
Level 6
Commissioning Handoff and Training

→

Level 7
Operational Readiness and Ongoing Commissioning (MBCx)

Level 5b turns operating margin from a design assumption into a validated boundary before production handoff.

Commissioning Stages



HIL Testing

Why Commissioning Should Not Stop After IST

What is Integrated Systems Testing

- IST is a Staged Maximum Load Test
- **PASS** **FAIL** Criteria are Based on Achieving Maximum Load Output & Maintaining Load through Equipment Failures
- Equipment and System Loop Tuning is Rarely Performed by or during IST.
- Most Temporary Equipment Used to Generate Load has Limitations



Why IST is Not Enough for Facility Operators

What Facility Operators Need

Reliable Thermal Management

- Tuned Equipment & Control Loops
- Smooth N+1 Transfers after Failures

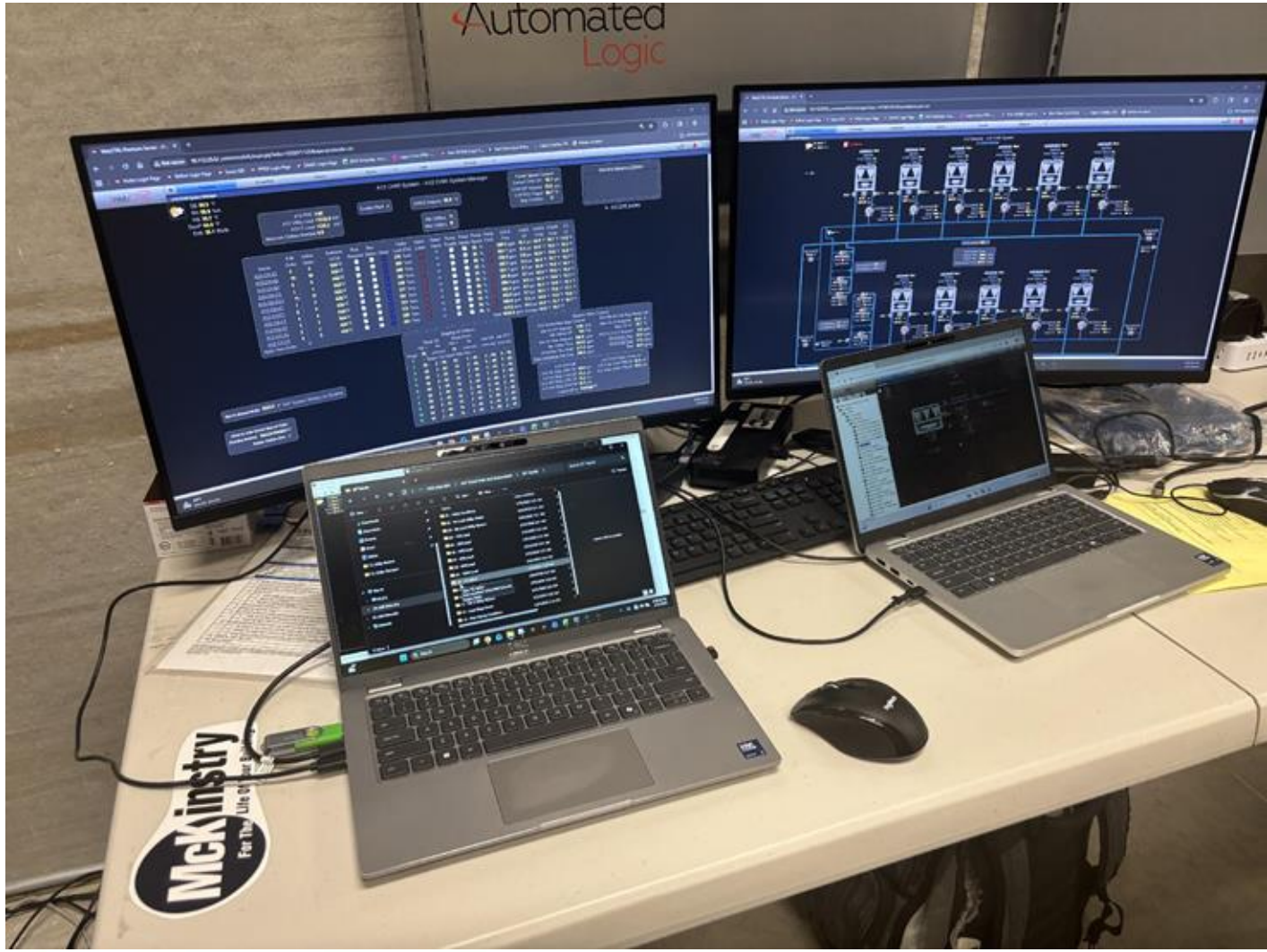
Efficient Heat Rejection

- Lowest Pressure and Flow Required for HX
- Lowest Total Facility PUE

Automated Identification of Performance Issues at Rack & Facility Level

- Visual Server Heat Map
- Cont. Automated Analysis of Available Data

**Using Design Model, Historical Sensor Data, and Current Performance Data with Digital Twin*



Liquid Cooled Load Banks (LCLBs) for Server Simulations

Example LCLBs (resistive)



Ambient Heat Rejection



CDU



Traditional

Typical heat profile: *steady state*



Boiler

steady state



Thermal Test Vehicle (TTV)

dynamic loading

AI Factory Considerations

- **Scalability –** Time constraints arise placing a load bank at every rack position. At massive scale, this is not feasible. Additionally, early procurement is critical.
- **Demarcation –** Increased complexity with stakeholders (liquid cooling alone includes responsibility of CDUs, TCS, flush fluid, coolant supplier, etc.).
- **CPU vs. GPU/TPU Commissioning –** Thermal loads in AI factories can be extreme. Testing typical load and upset conditions require proper rating selections up front.

validate operational integrity, redundancy, and thermal behavior before live GPU/TPU

Note: Air cooled load banks can be utilized as well onsite – the key focus of this paper is commissioning considerations in high-density AI applications requiring liquid cooling.

Liquid Cooled Load Bank (LCLB) Cleanliness

Assess as a 2-Prong Approach:

1. Pre-commissioning

- Disinfection cleaning prior to delivery
- Capped/sealed on both inlet and supply hookups
- Confirm wetted material compatibility
- Proper filtration selection

2. On-site Deployment

- Modular: Drain design of load bank to avoid contamination between jobs
- Potential flushing between systems
- Consistent filter cleanliness and efficiency
- Compliance with ISO 4406 (defines particle count limits for hydraulic fluids)

Overarching goal is to avoid contamination of the closed loop system, which can result in reduced heat transfer or premature failure during go-live

Performance Verification & Validation with Real Compute Equipment

Level 5b Comprehensive Test Matrix

1. Pre-Load Infrastructure Readiness

Direct-to-Chip (DTC) loops verified, commissioning filters swapped for permanent ones, and configuration frozen

2. AI Workload Power Validation

Real AI workloads run from Idle Baseline through Plan-of-Record (POR) validation down to minimum power caps.

3. Air & Liquid Flow PID Tuning

CDUs, fan and aircooling units tuned against live job launches using AI workload transients.

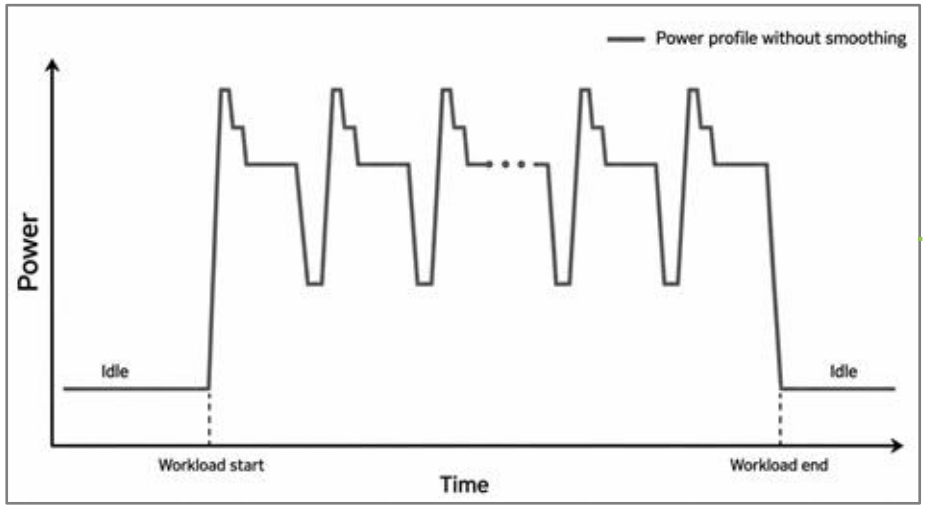
4. Dynamic Management & Acceptance

Validated across workload diversity bins and protected by automated safety fallback guards.

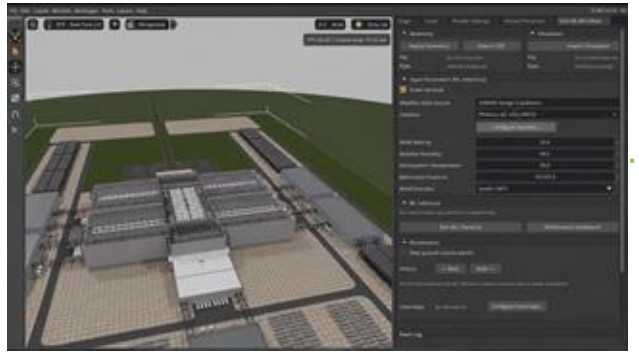
5. Digital Twin Calibration & Validation

Digital twin calibrated against measured power, thermal, and flow data to validate model accuracy and enable predictive operating scenarios.





CDU & Fans PID Tuning





Integrity, Safety, & Reliability Modules

The 5b/5c Bridge

1. **Define Operating Boundary**
2. **Establish Real-Time Workload Control**
3. **Execute Controlled Boundary Tests**
4. **Validate Protection & Recovery**
5. **Endurance Testing**

Safe Test Execution Framework

- **Real-time workload control —** Local authority to ramp, throttle, or stop compute.
- **Controlled test scope —** Limit the intended condition to the rack, pod, or module under test.
- **Coordinated execution —** Align IT, commissioning, controls/BMS, and facilities under one test director.
- **Recovery capacity —** Preserve cooling and pumping margin to restore normal operation.
- **Defined abort criteria —** Pre-set temperature, flow, power, and system limits before execution.

Validation Scenarios

High Temperature | Verify response near thermal boundary

Low Flow | Validate operation near hydraulic boundary

Alarm / Throttling | Confirm control-loop response

Recovery | Demonstrate return to stable operation

24–72 h Burn-In | Expose latent failures under sustained load

OCP GUIDANCE | **Boundary testing should be repeatable, observable, reversible, and executed with coordinated control of both compute load and cooling infrastructure.**

Boundary conditions should be introduced through controlled system/workload parameters; disruptive physical actions are outside the scope of this validation step.

Facility Operator Training & Handover - Level 6

Documentation

- Visual MEP & Control System One-Lines
- Digitally Synchronized Project Files with Maintenance System

Training

- Recorded Classroom & Field Walkthroughs by Engineer, Cx Provider, and MEP & Controls Representatives

Warranty & Transitional Support

- Ongoing Support with Technical Leads
- Regular Interval Operations Meetings Reviewing Performance Issues



Operationally Ready & Continuous MBCx - Level 7

Data Center Thermal Server Map

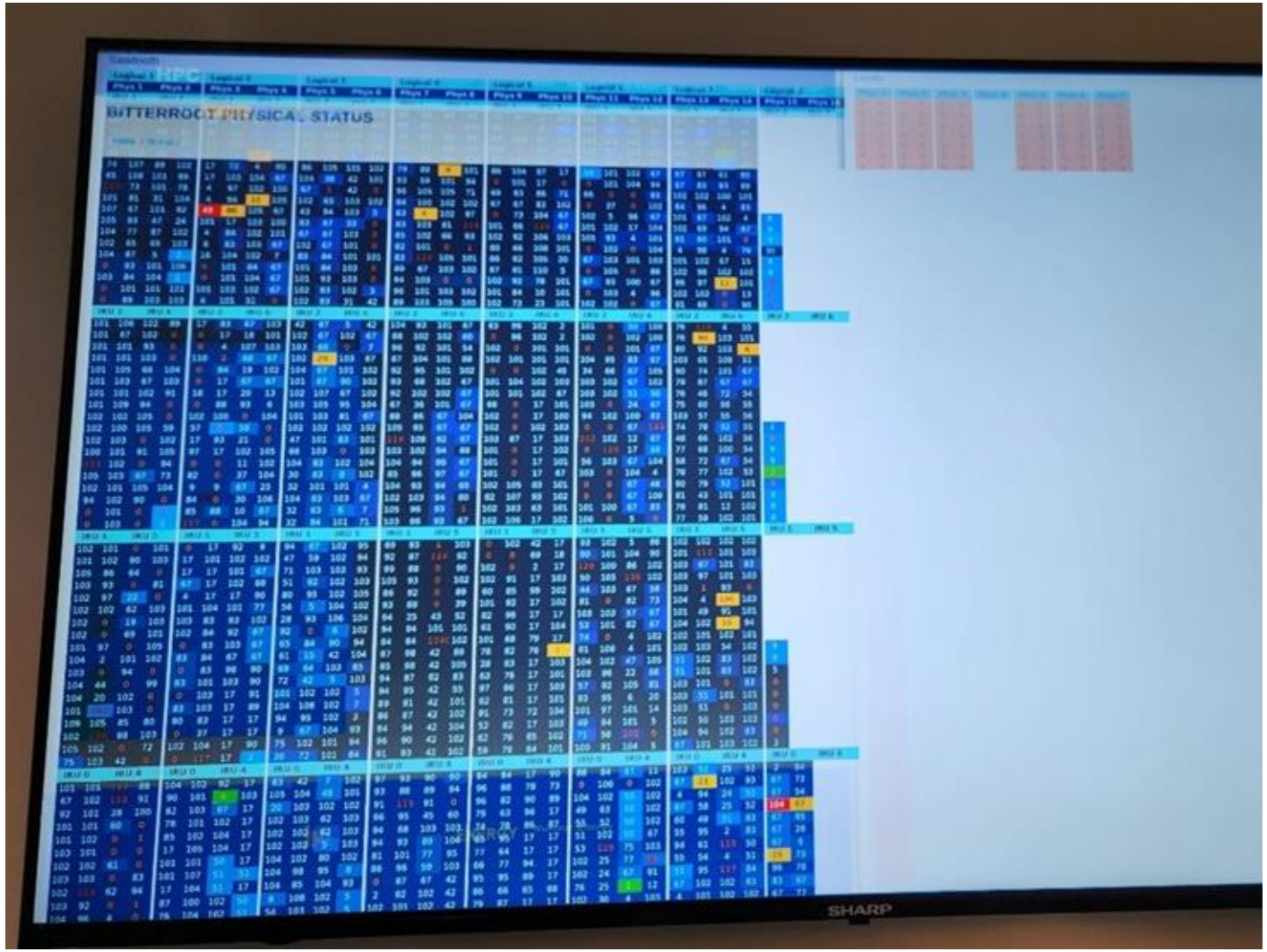
- Visual Display of Operating Temps

Automated Fault Detection & Response

- Management Systems Flags Hot Spots, Automated Adjusts Cooling Air Water Flows, and Rack Power Consumption to Maintain Operations
- Control System Shuts down Racks For Physical Inspection if Outside of Safe Operating Conditions

Digital Twins & Optimization Analysis

- Continuous Performance Data Review
- Validate Changes Before Deployment



Thank You!

